ClearCut – Intelligent Video Condenser

Sambaraju Saatwik Nandan
Computer Science and Engineering – Data Science
Gayatri Vidya Parishad College of Engineering (Autonomous)
Visakhapatnam, India.
322103383057@gvpce.ac.in

Singireddy Sai Lakshman Rao
Computer Science and Engineering – Data Science
Gayatri Vidya Parishad College of Engineering (Autonomous)
Visakhapatnam, India.
322103383059@gvpce.ac.in

Dr. G. V. Hindumathi
Computer Science and Engineering
Gayatri Vidya Parishad College of Engineering (Autonomous)
Visakhapatnam, India.
hindu@gvpce.ac.in

Mallela Amareswara Reddy
Computer Science and Engineering – Data Science
Gayatri Vidya Parishad College of Engineering (Autonomous)
Visakhapatnam, India.
322103383039@gvpce.ac.in

Pyla Jaya Venkata Sriram
Computer Science and Engineering – Data Science
Gayatri Vidya Parishad College of Engineering (Autonomous)
Visakhapatnam, India.
322103383052@gvpce.ac.in

Abstract— This paper presents an intelligent video condensation system powered by artificial intelligence for automatically summarizing long-form videos into concise and informative highlights. The system addresses the challenge of redundant and non-informative content commonly found in educational, entertainment, and sports videos by identifying and eliminating low-value segments such as silences, repetitions, and filler scenes. The proposed approach follows a two-stage summarization strategy. In the first stage, redundant frames are removed through integrated audio–visual analysis. In the second stage, scene-level semantic embeddings are employed to evaluate and rank video segments based on measures of novelty, diversity, and rarity, enabling the selection of the most significant portions. The proposed framework offers a scalable and efficient solution for automated video enhancement across multiple application domains.

Keywords: Video condensation, artificial intelligence, highlight extraction, audio-visual analysis, redundant frame removal, scene-level semantic embeddings, novelty detection, domain-aware summarization.

I. INTRODUCTION

The explosive growth of long-form video content across domains such as education, sports, entertainment, and surveillance has intensified the need for automatic video summarization systems that can distill lengthy recordings into compact, meaningful highlights. Manual summarization is inherently subjective and time-consuming, while naïve automatic approaches relying on uniform sampling or low-level heuristics often preserve redundant segments and overlook semantic structure. This challenge is particularly pronounced in videos with gradual scene transitions, recurring visual patterns, or prolonged static content, where redundancy dominates over novelty [1,2].

Early reinforcement learning–based methods framed video summarization as a sequential decision-making task. Approaches using diversity–representativeness reward functions [3] demonstrated improved temporal consistency and coverage of salient content, though they required careful reward engineering and significant computational resources. Sequence modeling and attention mechanisms further advanced the field. Cycle-consistent adversarial frameworks such as Cycle-SUM enforce summary fidelity through reconstruction constraints [4]. These methods improved semantic alignment but often depended on dense supervision or complex objectives, limiting cross-domain adaptability. Subsequent architectures incorporated spatio-temporal modeling, such as 3D U-Nets [6], enhancing the ability to capture long-range temporal dependencies, but these models also remained computationally intensive and less suited for real-time deployment.

More recent work has shifted toward unsupervised highlight detection that leverages intrinsic audio-visual recurrence [10]. By identifying segments that deviate from repetitive patterns, these methods isolate

semantically important events without labeled data, though they operate at fine temporal granularity and may struggle to maintain scene-level coherence.

Despite these advances, a gap remains between high-capacity summarization models and deployable systems that can robustly handle redundancy, preserve narrative continuity, and generalize across heterogeneous video types.

In this work, the proposed solution uses a comparatively lightweight, AI-powered video summarization framework that emphasizes scene-level semantic coherence and redundancy-aware selection. The system first reduces audio-visual redundancy by identifying repetitive or low-information segments and then performs scene-level semantic analysis using pretrained embeddings and similarity-based heuristics. This approach balances semantic richness with practical usability, enabling flexible deployment across educational, sports, and general-purpose videos.

II. LITERATURE SURVEY

Earlier work by Zhang et al. in "Cluster-based Video Summarization with Temporal Context Awareness" [1] combines visual clustering with temporal constraints to preserve narrative coherence. Although effective in maintaining continuity, the approach relies on handcrafted features and rigid clustering strategies, which limit semantic expressiveness. The absence of audio cues further reduces adaptability across diverse video domains.

In "Video Summarization with Attention-Based Encoder–Decoder Networks" [2], Ji et al. leverage attention mechanisms to predict frame importance within an encoder–decoder architecture. The approach improves saliency estimation by selectively focusing on informative frames. Despite this, it often relies on supervised or semi-supervised settings and operates at the frame level, which can result in fragmented summaries without higher-level temporal structure. Temporal context modeling remains relatively shallow unless combined with recurrent components.

In "Deep Reinforcement Learning for Unsupervised Video Summarization" [3], Zhou et al. employ reinforcement learning with diversity and representativeness rewards to generate summaries without supervision. While the framework effectively balances content coverage and redundancy reduction, RL training remains unstable and data-intensive. The approach is also limited to visual features and operates primarily at the frame level, which may compromise temporal coherence in the generated summaries.

Yuan et al. proposed Cycle-SUM [4], an unsupervised summarization approach based on cycle-consistent adversarial LSTM networks. The model enforces consistency between original videos and reconstructed summaries, enabling learning without annotations. However, adversarial training introduces instability and complexity, and the use of recurrent networks limits scalability and long-range temporal modeling. The method also focuses solely on visual information, excluding multimodal cues.

Several survey studies provide a structured understanding of the video summarization research landscape. Apostolidis et al. [5] offer a comprehensive taxonomy of deep learning–based video summarization methods, systematically categorizing supervised, unsupervised, and reinforcement learning approaches while highlighting emerging architectural trends.

Liu et al. introduced a reinforcement learning-based summarization framework in "Video Summarization through Reinforcement Learning with a 3D Spatio-Temporal U-Net" [6], where a 3D U-Net encodes spatio-temporal features and an RL agent learns frame-level keep or discard decisions. The method supports both supervised and unsupervised training modes and captures rich spatio-temporal information. Nevertheless, the frame-wise decision process can lead to temporal discontinuities such as jump cuts, and the combination of 3D CNNs with reinforcement learning results in high computational and memory requirements. Additionally, performance is sensitive to reward design and requires extensive training data.

Solbiati et al. explored unsupervised segmentation in "Unsupervised Topic Segmentation of Meetings with BERT Embeddings" [7], demonstrating that pretrained language models can effectively capture semantic transitions in long-form transcripts. However, the approach is restricted to textual data and does not address visual redundancy or multimodal video summarization, limiting its direct applicability to video content.

Meena et al. [8] examine both traditional and contemporary summarization techniques, emphasizing feature representations, evaluation protocols, and methodological evolution over

time.

Paramanantham et al. [9] further analyze key features and novel summarization methods, highlighting trends in deep learning–based approaches and evaluation strategies.

In "Unsupervised Video Highlight Detection by Learning from Audio and Visual Recurrence" [10], the authors propose an unsupervised highlight detection framework that exploits recurring audio and visual patterns across multiple related videos. By clustering videos into pseudo-categories and learning pseudo-highlight scores without annotations, the method demonstrates the effectiveness of cross-video recurrence cues. However, the approach assumes the availability of multiple similar videos, limiting its applicability to single-video summarization scenarios. Its reliance on clustering quality and lack of explicit temporal coherence modeling further restrict robustness, while the multi-video similarity computations introduce significant computational overhead.

Huynh-Lam et al. presented a domain-specific solution in "Context-Aware Surveillance Video Summarization" [11], prioritizing salient and abnormal activities in surveillance footage. While effective in structured monitoring environments, the method assumes predictable motion patterns and scene dynamics, significantly limiting generalization to open-domain videos such as lectures or sports.

Farhan and Hossain proposed context-aware temporal embeddings in [12], focusing on modeling object-level evolution and contextual changes across video sequences. While the learned embeddings improve semantic understanding of dynamic scenes, the work does not provide an end-to-end summarization pipeline. The object-centric design further increases computational complexity, limiting practical applicability.

In "A Framework for Video Summarization Using Visual Attention Technique" [13], Dhanushree et al. apply visual attention mechanisms to score frame saliency. While the method highlights visually prominent regions, it operates purely at the frame level and lacks redundancy reduction, temporal structuring, and multimodal integration, making it sensitive to visual noise.

Alaa et al. [14] further consolidate a wide range of summarization strategies, presenting an extensive comparative overview of models, datasets, and application domains.

Tian et al. introduced a transformer-based solution in "Video Summarization with Temporal-Channel Visual Transformer" [15], jointly modeling temporal and channel-wise dependencies to capture complex visual patterns. The method demonstrates strong representational capability for long-range dependencies. However, transformer architectures require large-scale training data and substantial computational resources, making them less suitable for lightweight or real-time deployment. The model also lacks explicit mechanisms for domain adaptation.

Vora et al. [16] provide a high-level discussion of AI-driven video summarization techniques for content retrieval and management. Although the work outlines potential applications of deep learning, it remains largely conceptual, lacking concrete system architectures, implementation details, and experimental validation.

III. PROPOSED SYSTEM

The proposed system is an AI-powered, general-purpose video summarization framework designed to generate concise and informative highlights from long-form videos across multiple domains. The system emphasizes practical applicability and flexibility, allowing domain-specific behavior through configurable parameters rather than relying on a single fixed model. Currently, the framework supports educational, sports, and general video content, with the ability to adapt summarization strategies based on domain characteristics.

The overall architecture follows a two-stage summarization pipeline. In the first stage, the system performs audio-visual redundancy removal to eliminate low-value segments such as prolonged silences, repetitive scenes, and visually static frames. Audio signals are analyzed to detect silence and low-energy regions, while visual similarity between consecutive frames or short segments is computed to identify redundant content. This stage significantly reduces the effective video length while preserving essential temporal structure, thereby improving efficiency and downstream processing quality.

In the second stage, the reduced video is segmented into scenes, and scene-level semantic embeddings are generated using pretrained deep learning models. These embeddings capture high-level semantic information without requiring task-

specific training. A similarity-based heuristic scoring mechanism is then applied to rank scenes based on criteria such as novelty, diversity, and rarity. By comparing embedding similarities across scenes, the system prioritizes segments that contribute unique or informative content to the summary while suppressing repetitive or semantically similar portions.

Finally, the selected high-scoring segments are assembled into a coherent summary constrained by a user-defined duration. Temporal ordering is preserved to maintain narrative flow, ensuring that the generated summary remains contextually understandable. The modular and lightweight nature of the proposed system makes it suitable for prototyping and real-world deployment, providing a scalable foundation for intelligent video summarization without the computational overhead of end-to-end deep learning models.

IV. METHODOLOGY

The proposed video summarization system adopts a multimodal and modular processing pipeline that integrates audio and visual cues to generate concise highlight reels. The methodology is structured to progressively eliminate redundancy, extract semantic information, and select informative segments in a domain-aware manner, as illustrated in the system flowchart in figure 1.

A. Input Processing

The system takes a raw video as input and separates it into its audio stream and visual frames. These modalities are processed in parallel to capture complementary information essential for identifying informative and redundant content.

B. Audio Processing Layer

The extracted audio is processed using an automatic speech transcription and embedding pipeline based on the Whisper model. This step converts spoken content into text and generates semantic embeddings that represent the contextual meaning of audio segments. Periods of silence and low-energy audio are implicitly identified during transcription and are treated as low-value regions. The resulting audio embeddings serve as semantic cues for downstream multimodal scoring.

C. Visual Processing Layer

The visual stream undergoes scene segmentation using PySceneDetect with decay-based thresholding to identify scene boundaries and structural changes in the video. Following segmentation, visually redundant frames and near-duplicate scenes are removed by computing similarity between consecutive frames or segments. This step significantly reduces repetitive content, especially in videos with static or slow-changing visuals, while preserving meaningful temporal structure.

D. Multimodal Feature Extraction and Scoring

Outputs from the audio and visual processing layers are fused in a multimodal feature extraction and scoring engine. Scene-level semantic embeddings derived from visual content are combined with audio embeddings to compute similarity-based scores. Each segment is evaluated based on heuristic criteria such as novelty, diversity, and semantic uniqueness relative to other segments. Domain-specific weighting parameters are applied at this stage to adapt the scoring behavior for different video types, such as educational or sports content.

E. Segment Ranking and Selection

Based on the computed scores, segments are ranked in descending order of importance. The system iteratively selects top-ranked segments while ensuring minimal semantic redundancy. The selection process continues until the accumulated duration satisfies a user-defined summary length, ensuring both relevance and conciseness.

F. Segment Stitching and Summary Generation

The selected segments are temporally ordered and stitched together using OpenCV and FFmpeg to produce the final summarized video. Temporal coherence is preserved to maintain narrative flow, resulting in a concise and informative highlight reel that reflects the most significant portions of the original video.

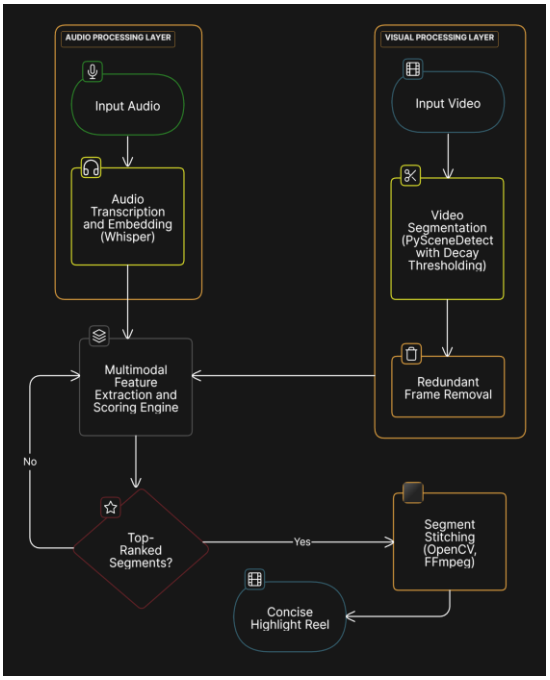

Figure 1: System architecture and processing pipeline of the proposed multimodal video summarization framework.

V. REFERENCES

[1] S. Zhang, Y. Zhu, and A. K. Roy-Chowdhury, "Cluster-based video summarization with temporal context awareness," IEEE Transactions on Image Processing, 2016. Available: https://vcg.ece.ucr.edu/sites/default/files/2019-02/tip2016_summarization.pdf

[2] Z. Ji, K. Xiong, Y. Pang, and X. Li, "Video summarization with attention-based encoder-decoder networks," arXiv:1708.09545, 2017. Available: https://arxiv.org/abs/1708.09545

[3] K. Zhou, Y. Qiao, and T. Xiang, "Deep reinforcement learning for unsupervised video summarization with diversity-representativeness reward," arXiv:1801.00054, 2018. Available: https://arxiv.org/abs/1801.00054

[4] L. Yuan, F. E. H. Tay, P. Li, L. Zhou, and J. Feng, "Cycle-SUM: Cycle-consistent adversarial LSTM networks for unsupervised video summarization," arXiv:1904.08265, 2019. Available: https://arxiv.org/abs/1904.08265

[5] E. Apostolidis et al., "Video summarization using deep neural networks: A survey," arXiv:2101.06072, 2021. Available: https://arxiv.org/abs/2101.06072

[6] T. Liu et al., "Video summarization through reinforcement learning with a 3D spatio-temporal U-Net," arXiv:2106.10528, 2021. Available: https://arxiv.org/abs/2106.10528

[7] A. Solbiati et al., "Unsupervised topic segmentation of meetings with BERT embeddings," arXiv:2106.12978, 2021. Available: https://arxiv.org/abs/2106.12978

[8] P. Meena, H. Kumar, and S. Yadav, "A review on video summarization techniques," Engineering Applications of Artificial Intelligence, 2023. Available: https://www.sciencedirect.com/science/article/abs/pii/S0952197622006571

[9] V. Paramanantham and S. S. Kumar, "A review on key features and novel methods for video summarization," International Journal of Engineering and Advanced Technology, 2023. Available: https://www.ijeat.org/wp-

content/uploads/papers/v12i3/F37370811622.pdf

[10] Z. Islam, S. Paul, M. Rochan, "Unsupervised video highlight detection by learning from audio and visual recurrence," arXiv:2407.13933, 2024. Available: https://arxiv.org/abs/2407.13933

[11] H.-D. Huynh-Lam, N.-P. Ho-Thi, M.-T. Tran, and T.-N. Le, "Context-aware surveillance video summarization," IEEE Transactions on Image Processing, 2024. Available: https://arxiv.org/abs/2404.04511

[12] A. Farhan and M. S. Hossain, "Context-aware temporal embedding of objects in video data," arXiv:2408.12789, 2024. Available: https://arxiv.org/abs/2408.12789

[13] M. Dhanushree, R. Priya, P. Aruna, and R. Bhavani, "A framework for video summarization using visual attention technique," Indian Journal of Science and Technology, 2024. Available: https://indjst.org/articles/a-framework-for-video-summarization-using-visual-attention-technique

[14] T. Alaa et al., "Video summarization techniques: A comprehensive review," Proceedings of the International Conference on Data Science, 2024. Available: https://www.scitepress.org/Papers/2024/129364/129364.pdf

[15] X. Tian, J. Jin, and X. Tang, "Video summarization with temporal-channel visual transformer," Pattern Recognition, 2025. Available: https://www.sciencedirect.com/science/article/pii/S0031320325002912

[16] D. Vora et al., "AI-driven video summarization for optimizing content retrieval and management through deep learning techniques," Scientific Reports, 2025. Available: www.nature.com/articles/s41598-025-87824-9